Power Creep In



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Final Project Write Up





Introduction

The Pokemon Trading Card Game, also known as Pokemon TCG for short, has been around since 1999. With the Pokemon video games being so popular, the TCG also grew immensely. Soon enough, there were regional and international tournaments. People are hooked onto the addictive nature of collecting and battling Pokemon.

However, Pokemon TCG has been falling into a common game design trap that many other trading card and video games are known for called power creep. As more expansions and cards come out, the more powerful these new cards become. More health, more damage, powerful abilities, each new expansion would come with an incentive to buy and go for these new cards. This issue would make cards from the past so irrelevant that even official, competitive Pokemon TCG rules would adapt a cycle format where only cards found from the latest set could be used for tournament play.



How Bad Is Power Creep?

Hitting 25 years old this year, Pokemon TCG is growing fast. As new cards and mechanics are brought in and out of the competitive meta, let's take a step back and see just how powerful these cards have gotten.

We'll be examining the most powerful cards that were released every year, starting from 1999 all the way to 2024. How strong the card is will simply be determined by how large their hit points (HP) are, basically how much health a card has. The cards will also be split by evolution stage. There are three evolution stages: Basic, Stage 1, and Stage 2.

Three datasets will be used. A CSV file containing information of all cards released from 1999 - 2023 and an API that contains the latest card releases of this year, allowing us to gain a full picture of these card's evolutions throughout the years. We will also use a CSV containing information about the set that these cards were released in and see how powerful each set of cards were.



The Process

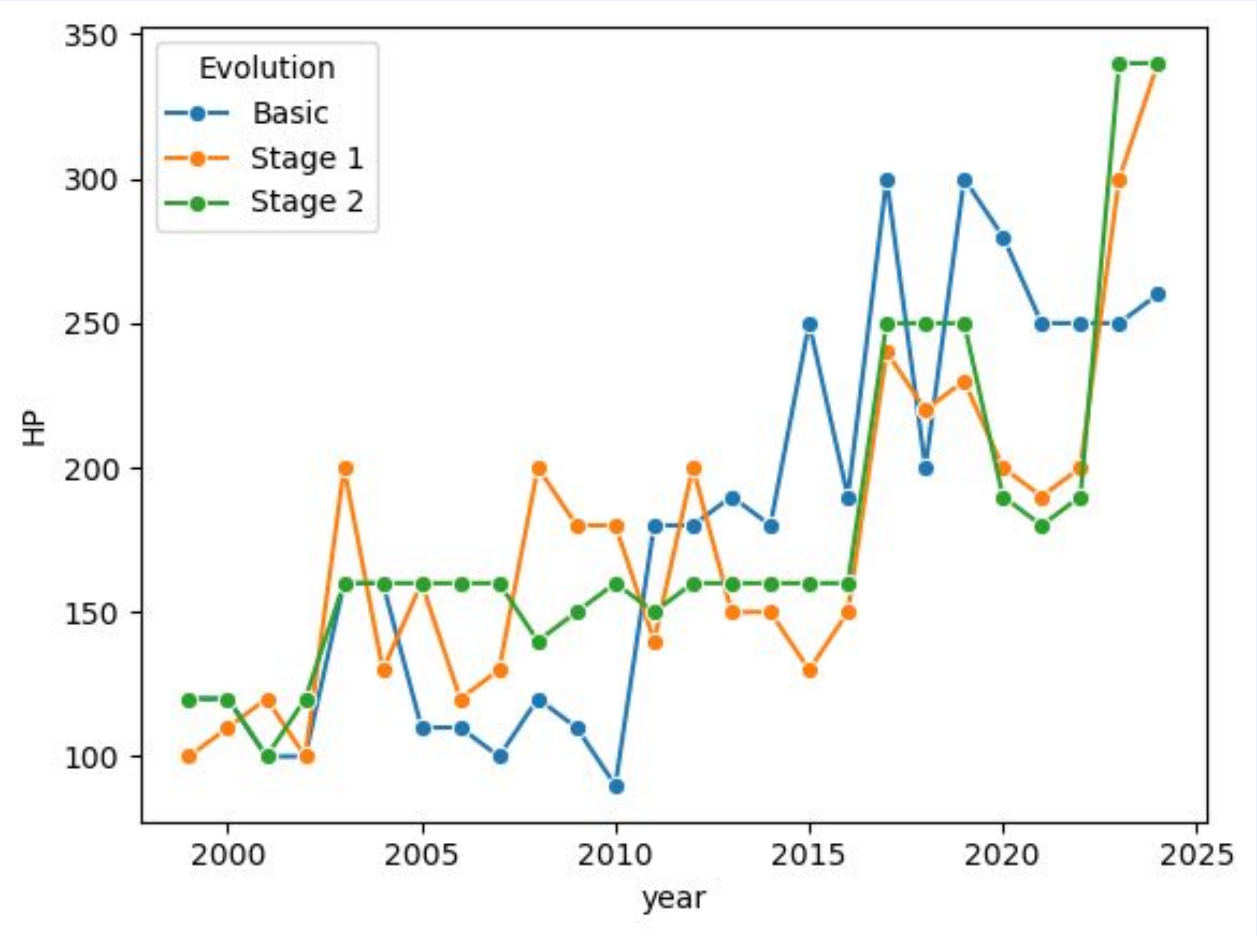
Like mentioned before, I had three sets of data. The card data CSV and API would work together will the set data CSV would be its own individual process. With the card data the process was to collect the necessary data from both of them (Name, Year Release, HP, Evolutionary Stage), and mash them into one dataframe that I would be able to create a cohesive graph with.

The CSV was quite easy to work with, Pandas was able to quickly make a dataframe where the only challenge was to separate the base evolutionary stage from the “subtypes” category as Pokemon can have more than one subtype. For example you can have a basic EX and so forth

The API was a little more difficult as this one was much more complex and expansive, with endpoints and page flipping being my worst enemy. Once I was able to navigate through it was just like the same process as the CSV. After they were both cleaned and ready, I combined them and my first visualization.

As for the second graph, the set data was simple to clean and get ready for the plot, it was making the plot that was the most difficult part. There just were too many sets for that one small graph to handle, despite my best effort to even trim the fat of some promotional sets, I will say I’m not very happy with how it turned out and I hope to do better next time. But it's all about learning and getting better.

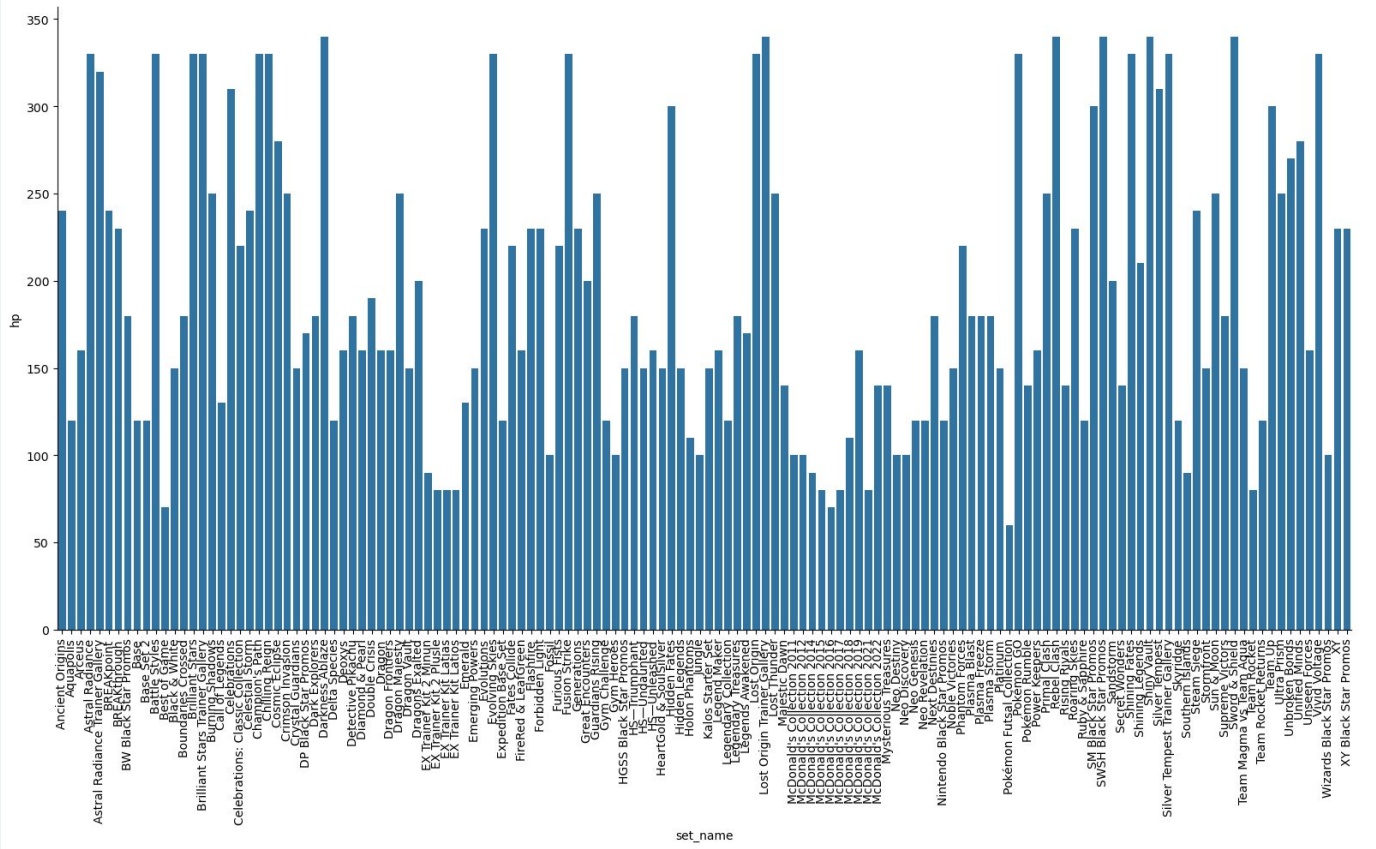
The Results



One look at this graph and its clear just how bad power creep has gotten in Pokemon TCG. The most powerful Basic Pokemon of this year more than doubles the most powerful Stage 2 Pokemon of 1999. Both Stage 1 and 2 Pokemon have reached almost 350 HP.

Although you can notice that there might have been an attempt to cut down and lower the power of cards between 2020 and 2023. This was during the Sword and Shield expansion, something that wasn't too popular amongst players, maybe that's why they doubled down and went even higher HP in the 2024, especially with the introduction of a new mechanic called Terra Pokemon.

The Results



I understand it's a hard graph to read but allow me to help guide you through it. You'll notice a big inconsistency in power between sets. Most of the lowered powered sets are from promotional packs with McDonalds or other companies.

The ones that are higher are usually tied in with a Pokemon video game coming out around the same time. It's clear how the Pokemon Company carefully select which sets are going to be more powerful based off what could get more of their property sold.